US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12403055
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Hight; Joshua C. et al.

Surgical patient support for lateral-to-prone patient positioning

Abstract

According to the present disclosure, a surgical patient support provides support to a patient. The surgical patient support may include configuration to accommodate various patient body positions to provide a variety of access to the patient's body.

Inventors: **Hight; Joshua C. (Littleton, MA), Drake; Jesse S. (Westborough, MA), Marrion; Jeffrey C. (Acton, MA), Dubois; Christopher B. (Marlborough, MA), Hertz; Ben (Acton, MA), Sennett; Andrew (Hanover, MA), Brophy; Patrick (Medford, MA)**

Applicant: **Allen Medical Systems, Inc.** (Batesville, IN)

Family ID: **57178326**

Assignee: **Allen Medical Systems, Inc. (Batesville, IN)**

Appl. No.: **17/386731**

Filed: **July 28, 2021**

Prior Publication Data

Document Identifier	Publication Date
US 20210353487 A1	Nov. 18, 2021

Related U.S. Application Data

continuation parent-doc US 16451446 20190625 US 11096853 child-doc US 17386731
continuation parent-doc US 15290156 20161011 US 10363189 20190730 child-doc US 16451446
us-provisional-application US 62352711 20160621
us-provisional-application US 62245646 20151023

Publication Classification

Int. Cl.: **A61G13/08** (20060101); **A61G13/00** (20060101); **A61G13/04** (20060101); **A61G13/06** (20060101); **A61G13/12** (20060101)

U.S. Cl.:

CPC **A61G13/08** (20130101); **A61G13/0054** (20161101); **A61G13/04** (20130101); **A61G13/1245** (20130101); **A61G13/1295** (20130101); A61G13/06 (20130101); A61G13/122 (20130101); A61G13/123 (20130101); A61G13/1235 (20130101); A61G2200/322 (20130101); A61G2200/325 (20130101)

Field of Classification Search

CPC: A61G (13/04); A61G (13/06); A61G (13/08); A61G (13/0054); A61G (13/122); A61G (13/123); A61G (13/1235); A61G (13/1245); A61G (13/1295); A61G (2200/322); A61G (2200/325)

USPC: 5/613

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
866309	12/1906	Scanlan	N/A	N/A
1021335	12/1911	Robinson et al.	N/A	N/A
1098477	12/1913	Cashman	N/A	N/A
1160451	12/1914	Sanford	N/A	N/A
1171713	12/1915	Gilkerson	N/A	N/A
1372565	12/1920	Skelly	N/A	N/A
1528835	12/1924	McCollough	N/A	N/A
1662464	12/1927	McCutchen	N/A	N/A
1799692	12/1930	Knott	N/A	N/A
1938006	12/1932	Blanchard	N/A	N/A
2103693	12/1936	Pohl	N/A	N/A
2188592	12/1939	Hosken	N/A	N/A
2261297	12/1940	Seib	N/A	N/A
2337505	12/1942	Swift	N/A	N/A
2452816	12/1947	Wagner	N/A	N/A
2613371	12/1951	Keyes, Jr.	N/A	N/A
2636793	12/1952	Meyer	N/A	N/A
2667169	12/1953	Kambourakis	N/A	N/A
2688410	12/1953	Nelson	N/A	N/A
2691979	12/1953	Watson	N/A	N/A
2764150	12/1955	Ettinger et al.	N/A	N/A
2792945	12/1956	Brenny	N/A	N/A
2803022	12/1956	Wynkoop	N/A	N/A
2880720	12/1958	Houghtaling	N/A	N/A
3046071	12/1961	Shampaine et al.	N/A	N/A
3049726	12/1961	Getz	N/A	N/A
3090381	12/1962	Watson	N/A	N/A
3206188	12/1964	Douglass, Jr.	N/A	N/A

3226734	12/1965	Coventon	N/A	N/A
3238539	12/1965	Albert	N/A	N/A
3281141	12/1965	Smiley	N/A	N/A
3286707	12/1965	Shafer	N/A	N/A
3302218	12/1966	Stryker	N/A	N/A
3388700	12/1967	Mountz	N/A	N/A
3428307	12/1968	Owen	N/A	N/A
3434165	12/1968	Keane	N/A	N/A
3584321	12/1970	Buchanan	N/A	N/A
3599964	12/1970	Magni	N/A	N/A
3640416	12/1971	Temple	N/A	N/A
3652851	12/1971	Zaalberg	N/A	N/A
3739406	12/1972	Koetter	N/A	N/A
3745996	12/1972	Rush	N/A	N/A
3751028	12/1972	Scheininger et al.	N/A	N/A
3766384	12/1972	Anderson et al.	N/A	N/A
3795018	12/1973	Broaded	N/A	N/A
3814414	12/1973	Chapa	N/A	N/A
3827089	12/1973	Grow	N/A	N/A
3832742	12/1973	Stryker	N/A	N/A
3859982	12/1974	Dove	N/A	N/A
3873081	12/1974	Smith et al.	N/A	N/A
3895403	12/1974	Davis et al.	N/A	N/A
3946452	12/1975	Eary	N/A	N/A
3949983	12/1975	Tommasino et al.	N/A	N/A
3988790	12/1975	Mracek et al.	N/A	N/A
4071916	12/1977	Nelson	N/A	N/A
4101120	12/1977	Seshima	N/A	N/A
4131802	12/1977	Braden et al.	N/A	N/A
4144880	12/1978	Daniels	N/A	N/A
4148472	12/1978	Rais	N/A	N/A
4175550	12/1978	Leininger et al.	N/A	N/A
4186917	12/1979	Rais et al.	N/A	N/A
4227269	12/1979	Johnston	N/A	N/A
4239039	12/1979	Thompson	N/A	N/A
4244358	12/1980	Pyers	N/A	N/A
4257407	12/1980	Macchi	N/A	N/A
4356577	12/1981	Taylor et al.	N/A	N/A
4384378	12/1982	Getz et al.	N/A	N/A
4398707	12/1982	Cloward	N/A	N/A
4459712	12/1983	Pathan	N/A	N/A
4503844	12/1984	Siczek et al.	N/A	N/A
4545571	12/1984	Chambron	N/A	N/A
4552346	12/1984	Schnelle et al.	N/A	N/A
4579111	12/1985	Ledesma	N/A	N/A
4658450	12/1986	Thompson	N/A	N/A
4678171	12/1986	Sanders	N/A	N/A
4712781	12/1986	Watanabe	N/A	N/A
4730606	12/1987	Leininger	N/A	N/A
4763643	12/1987	Vrzalik	N/A	N/A

4769584	12/1987	Irigoyen et al.	N/A	N/A
4771785	12/1987	Duer et al.	N/A	N/A
4827541	12/1988	Vollman	N/A	N/A
4840362	12/1988	Bremer et al.	N/A	N/A
4850775	12/1988	Lee et al.	N/A	N/A
4858128	12/1988	Alsip et al.	N/A	N/A
4866796	12/1988	Robinson et al.	N/A	N/A
4868937	12/1988	Connolly	N/A	N/A
4872657	12/1988	Lussi	N/A	N/A
4887325	12/1988	Tesch	N/A	N/A
4924537	12/1989	Alsip et al.	N/A	N/A
4937901	12/1989	Brennan	N/A	N/A
4939801	12/1989	Schaal et al.	N/A	N/A
4944054	12/1989	Bossert	N/A	N/A
4944500	12/1989	Mueller et al.	N/A	N/A
4947496	12/1989	Connolly	N/A	N/A
4953245	12/1989	Jung	N/A	N/A
4970737	12/1989	Sagel	N/A	N/A
5020170	12/1990	Ruf	N/A	N/A
5088706	12/1991	Jackson	N/A	N/A
5131106	12/1991	Jackson	N/A	N/A
5152024	12/1991	Chrones	N/A	N/A
5161267	12/1991	Smith	N/A	N/A
5181289	12/1992	Kassai	N/A	N/A
5210887	12/1992	Kershaw	N/A	N/A
5210888	12/1992	Canfield	N/A	N/A
5231741	12/1992	Maguire	N/A	N/A
5239716	12/1992	Fisk	N/A	N/A
5274862	12/1993	Palmer et al.	N/A	N/A
5333334	12/1993	Kassai	N/A	N/A
5393018	12/1994	Roth et al.	N/A	N/A
5404603	12/1994	Fukai et al.	N/A	N/A
5444882	12/1994	Andrews et al.	N/A	N/A
5461740	12/1994	Pearson	5/607	A61G 7/012
5483323	12/1995	Matsuda et al.	N/A	N/A
5487195	12/1995	Ray	N/A	N/A
5499409	12/1995	Nix	N/A	N/A
5502853	12/1995	Singleton et al.	N/A	N/A
5524304	12/1995	Shutes	N/A	N/A
5544371	12/1995	Fuller	N/A	N/A
5579550	12/1995	Bathrick et al.	N/A	N/A
5588705	12/1995	Chang	N/A	N/A
5613254	12/1996	Clayman et al.	N/A	N/A
5658315	12/1996	Lamb et al.	N/A	N/A
5673443	12/1996	Marmor	N/A	N/A
5737781	12/1997	Votel et al.	N/A	N/A
5775334	12/1997	Lamb et al.	N/A	N/A
5778467	12/1997	Scott et al.	N/A	N/A
5794286	12/1997	Scott et al.	N/A	N/A

5890238	12/1998	Votel et al.	N/A	N/A
5901388	12/1998	Cowan	N/A	N/A
5926871	12/1998	Howard	N/A	N/A
5937456	12/1998	Norris	N/A	N/A
5950259	12/1998	Boggs	N/A	N/A
6003174	12/1998	Kantrowitz et al.	N/A	N/A
6035465	12/1999	Rogozinski	N/A	N/A
6042558	12/1999	Hoyne et al.	N/A	N/A
6049923	12/1999	Ochiai	N/A	N/A
6076525	12/1999	Hoffman	N/A	N/A
6094760	12/1999	Nonaka et al.	N/A	N/A
6108838	12/1999	Connolly et al.	N/A	N/A
6112349	12/1999	Connolly	N/A	N/A
6154901	12/1999	Carr	N/A	N/A
6230342	12/2000	Haugs	N/A	N/A
6260220	12/2000	Lamb et al.	N/A	N/A
6282736	12/2000	Hand	N/A	N/A
6286164	12/2000	Lamb et al.	N/A	N/A
6295671	12/2000	Reesby et al.	N/A	N/A
6311349	12/2000	Kazakia et al.	N/A	N/A
6315564	12/2000	Levisman	N/A	N/A
6324710	12/2000	Hernandez et al.	N/A	N/A
6385801	12/2001	Watanabe et al.	N/A	N/A
6421854	12/2001	Heimbrock	N/A	N/A
6438777	12/2001	Bender	N/A	N/A
6496991	12/2001	Votel	N/A	N/A
6499158	12/2001	Easterling	N/A	N/A
6499160	12/2001	Hand et al.	N/A	N/A
6505365	12/2002	Hanson et al.	N/A	N/A
6523197	12/2002	Zitzmann	N/A	N/A
6526610	12/2002	Hand	N/A	N/A
6584630	12/2002	Dinkler	N/A	N/A
6609260	12/2002	Hand et al.	N/A	N/A
6615430	12/2002	Heimbrock	N/A	N/A
6622324	12/2002	VanSteenburg et al.	N/A	N/A
6634043	12/2002	Lamb et al.	N/A	N/A
6637058	12/2002	Lamb	N/A	N/A
6638299	12/2002	Cox	N/A	N/A
6662388	12/2002	Friel et al.	N/A	N/A
6662391	12/2002	Wilson et al.	N/A	N/A
6668396	12/2002	Wei	N/A	N/A
6671904	12/2003	Easterling	N/A	N/A
6681423	12/2003	Zachrisson	N/A	N/A
6691347	12/2003	Hand et al.	N/A	N/A
6701553	12/2003	Hand	N/A	N/A
6701554	12/2003	Heimbrock	N/A	N/A
6721976	12/2003	Schwaegerle	N/A	N/A
6813788	12/2003	Dinkler et al.	N/A	N/A
6817363	12/2003	Biondo et al.	N/A	N/A
6854137	12/2004	Johnson	N/A	N/A

6857144	12/2004	Huang	N/A	N/A
6859967	12/2004	Harrison et al.	N/A	N/A
6862759	12/2004	Hand et al.	N/A	N/A
6862761	12/2004	Hand et al.	N/A	N/A
6874181	12/2004	Vijayendran	N/A	N/A
6886199	12/2004	Schwaegerle	N/A	N/A
6898811	12/2004	Zucker et al.	N/A	N/A
6912959	12/2004	Kolody et al.	N/A	N/A
6928676	12/2004	Schwaegerle	N/A	N/A
6941951	12/2004	Hubert et al.	N/A	N/A
6966081	12/2004	Sharps et al.	N/A	N/A
6971997	12/2004	Ryan et al.	N/A	N/A
6986179	12/2005	Varadharajulu et al.	N/A	N/A
7020917	12/2005	Kolody et al.	N/A	N/A
7080422	12/2005	Ben-Levi	N/A	N/A
7086103	12/2005	Barthelt	N/A	N/A
7089612	12/2005	Rocher et al.	N/A	N/A
7089884	12/2005	Wang et al.	N/A	N/A
7103932	12/2005	Kandora	N/A	N/A
7137160	12/2005	Hand et al.	N/A	N/A
7152261	12/2005	Jackson	N/A	N/A
7171709	12/2006	Weismiller	N/A	N/A
7197778	12/2006	Sharps	N/A	N/A
7214138	12/2006	Stivers et al.	N/A	N/A
7216385	12/2006	Hill	N/A	N/A
7234180	12/2006	Horton et al.	N/A	N/A
7290302	12/2006	Sharps	N/A	N/A
7343635	12/2007	Jackson	N/A	N/A
7343916	12/2007	Biondo et al.	N/A	N/A
7496980	12/2008	Sharps	N/A	N/A
7520007	12/2008	Skripps	N/A	N/A
7520008	12/2008	Wong et al.	N/A	N/A
7565708	12/2008	Jackson	N/A	N/A
7600281	12/2008	Skripps	N/A	N/A
7653953	12/2009	Lopez-Sansalvador et al.	N/A	N/A
7669262	12/2009	Skripps	N/A	N/A
7681269	12/2009	Biggie et al.	N/A	N/A
7694369	12/2009	Hinders et al.	N/A	N/A
7739762	12/2009	Lamb	N/A	N/A
7810185	12/2009	Burstner et al.	N/A	N/A
7824353	12/2009	Matta	N/A	N/A
7861720	12/2010	Wolcott	N/A	N/A
7882583	12/2010	Skripps	N/A	N/A
7931607	12/2010	Biondo et al.	N/A	N/A
7954996	12/2010	Boomgaarden et al.	N/A	N/A
D645967	12/2010	Sharps	N/A	N/A
8020227	12/2010	Dimmer et al.	N/A	N/A
8042208	12/2010	Gilbert et al.	N/A	N/A
8056163	12/2010	Lemire	N/A	N/A

8060960	12/2010	Jackson	N/A	N/A
8118029	12/2011	Gneiting et al.	N/A	N/A
D663427	12/2011	Sharps	N/A	N/A
D665912	12/2011	Skripps	N/A	N/A
8234730	12/2011	Skripps	N/A	N/A
8234731	12/2011	Skripps	N/A	N/A
8256050	12/2011	Wong et al.	N/A	N/A
8286283	12/2011	Copeland et al.	N/A	N/A
D676971	12/2012	Sharps	N/A	N/A
8381331	12/2012	Sharps et al.	N/A	N/A
8397323	12/2012	Skripps et al.	N/A	N/A
D683032	12/2012	Sharps	N/A	N/A
8464375	12/2012	Jorgensen et al.	N/A	N/A
8486068	12/2012	Starr	N/A	N/A
8555439	12/2012	Soto et al.	N/A	N/A
8584281	12/2012	Diel et al.	N/A	N/A
8590074	12/2012	Hornbach et al.	N/A	N/A
8635725	12/2013	Tannoury et al.	N/A	N/A
8676293	12/2013	Breen et al.	N/A	N/A
8677529	12/2013	Jackson	N/A	N/A
8707476	12/2013	Sharps	N/A	N/A
8707484	12/2013	Jackson et al.	N/A	N/A
8719979	12/2013	Jackson	N/A	N/A
8732876	12/2013	Lachenbruch et al.	N/A	N/A
8763178	12/2013	Martin et al.	N/A	N/A
8777878	12/2013	Deitz	N/A	N/A
8782832	12/2013	Blyakher et al.	N/A	N/A
8806679	12/2013	Soto et al.	N/A	N/A
8826474	12/2013	Jackson	N/A	N/A
8826475	12/2013	Jackson	N/A	N/A
8833707	12/2013	Steinberg et al.	N/A	N/A
8839471	12/2013	Jackson	N/A	N/A
8844077	12/2013	Jackson et al.	N/A	N/A
8845264	12/2013	Kubiak et al.	N/A	N/A
8856986	12/2013	Jackson	N/A	N/A
8864205	12/2013	Lemire et al.	N/A	N/A
8893333	12/2013	Soto et al.	N/A	N/A
D720076	12/2013	Sharps et al.	N/A	N/A
8938826	12/2014	Jackson	N/A	N/A
8978180	12/2014	Jackson	N/A	N/A
8997286	12/2014	Wyslucha et al.	N/A	N/A
9119610	12/2014	Matta et al.	N/A	N/A
9180062	12/2014	Jackson	N/A	N/A
9205013	12/2014	Jackson	N/A	N/A
9211223	12/2014	Jackson	N/A	N/A
9233037	12/2015	Sharps et al.	N/A	N/A
9289342	12/2015	Jackson	N/A	N/A
9295433	12/2015	Jackson et al.	N/A	N/A
9308145	12/2015	Jackson	N/A	N/A
9339430	12/2015	Jackson et al.	N/A	N/A

9364380	12/2015	Jackson	N/A	N/A
9498397	12/2015	Hight et al.	N/A	N/A
10363189	12/2018	Hight et al.	N/A	N/A
11096853	12/2020	Hight et al.	N/A	N/A
2002/0138903	12/2001	Simmons et al.	N/A	N/A
2004/0133983	12/2003	Newkirk et al.	N/A	N/A
2005/0155149	12/2004	Pedersen	N/A	N/A
2005/0235415	12/2004	Pedersen et al.	N/A	N/A
2006/0123552	12/2005	Ben-Levi	N/A	N/A
2008/0000028	12/2007	Lemire et al.	N/A	N/A
2009/0044813	12/2008	Gneiting et al.	N/A	N/A
2009/0126116	12/2008	Lamb et al.	N/A	N/A
2009/0205139	12/2008	Van Deursen et al.	N/A	N/A
2009/0282614	12/2008	Jackson	N/A	N/A
2010/0192300	12/2009	Tannoury et al.	N/A	N/A
2011/0107516	12/2010	Jackson	5/608	A61G 13/04
2012/0144589	12/2011	Skripps	N/A	N/A
2012/0198625	12/2011	Jackson	N/A	N/A
2013/0111666	12/2012	Jackson	N/A	N/A
2013/0205500	12/2012	Jackson	N/A	N/A
2013/0219623	12/2012	Jackson	N/A	N/A
2013/0254995	12/2012	Jackson	N/A	N/A
2013/0254996	12/2012	Jackson	N/A	N/A
2013/0254997	12/2012	Jackson	N/A	N/A
2013/0312181	12/2012	Jackson et al.	N/A	N/A
2013/0312187	12/2012	Jackson	N/A	N/A
2013/0312188	12/2012	Jackson	N/A	N/A
2013/0326813	12/2012	Jackson	N/A	N/A
2014/0007349	12/2013	Jackson	N/A	N/A
2014/0020181	12/2013	Jackson	N/A	N/A
2014/0033436	12/2013	Jackson	N/A	N/A
2014/0068861	12/2013	Jackson et al.	N/A	N/A
2014/0082842	12/2013	Jackson	N/A	N/A
2014/0109316	12/2013	Jackson et al.	N/A	N/A
2014/0173826	12/2013	Jackson	N/A	N/A
2014/0196212	12/2013	Jackson	N/A	N/A
2014/0201913	12/2013	Jackson	N/A	N/A
2014/0201914	12/2013	Jackson	N/A	N/A
2014/0208512	12/2013	Jackson	N/A	N/A
2014/0317847	12/2013	Jackson	N/A	N/A
2014/0325759	12/2013	Bly	N/A	N/A
2015/0059094	12/2014	Jackson	N/A	N/A
2015/0150743	12/2014	Jackson	N/A	N/A
2015/0283017	12/2014	Harris, Jr.	N/A	N/A
2016/0000621	12/2015	Jackson et al.	N/A	N/A
2016/0000626	12/2015	Jackson et al.	N/A	N/A
2016/0000627	12/2015	Jackson et al.	N/A	N/A
2016/0000629	12/2015	Jackson et al.	N/A	N/A
2016/0361218	12/2015	Dubois et al.	N/A	N/A

2017/0112699	12/2016	Hight et al.	N/A	N/A
2019/0314236	12/2018	Hight et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1162508	12/1966	DE	N/A
3438956	12/1984	DE	N/A
4039907	12/1990	DE	N/A
4429062	12/1995	DE	N/A
202008001952	12/1999	DE	N/A
19723927	12/2002	DE	N/A
10158470	12/2002	DE	N/A
0501712	12/1991	EP	N/A
0617947	12/1994	EP	N/A
1210049	12/2001	EP	N/A
1686944	12/2005	EP	N/A
1159947	12/2005	EP	N/A
1982680	12/2007	EP	N/A
2247194	12/1974	FR	N/A
2210554	12/1988	GB	N/A
2001-112582	12/2000	JP	N/A
2004/026212	12/2003	WO	N/A
2006/006106	12/2005	WO	N/A
2006/061606	12/2005	WO	N/A
2009/054969	12/2008	WO	N/A
WO 2009/071787	12/2008	WO	N/A
2011/162803	12/2010	WO	N/A

OTHER PUBLICATIONS

Japanese Office Action for Japanese Patent Application No. 2016-207199 dated Apr. 3, 2018 and its English translation (15 pages). cited by applicant
Extended European Search Report for European Patent Application No. 16194763.5 dated Mar. 22, 2017 (7 pages). cited by applicant
Extended European Search Report for European Patent Application No. 18196057.6 dated Dec. 19, 2018 (7 pages). cited by applicant

---

Primary Examiner: Hare; David R

Attorney, Agent or Firm: Barnes & Thornburg LLP

Background/Summary

(1) The present application is a continuation of U.S. application Ser. No. 16/451,446, filed Jun. 25, 2019, now U.S. Pat. No. 11,096,853, which is a continuation of U.S. application Ser. No. 15/290,156, filed Oct. 11, 2016, now U.S. Pat. No. 10,363,189, which claims the benefit, under 35 U.S.C. § 119 (e), of U.S. Provisional Application No. 62/352,711, filed Jun. 21, 2016, and of U.S. Provisional Application No. 62/245,646, filed Oct. 23, 2015, and each of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

(1) The present disclosure relates to patient support devices and methods of operating patient support devices. More specifically, the present disclosure relates to surgical patient supports and methods of operating surgical patient supports.

(2) Patient supports devices, for example, those of surgical patient supports can provide support to patient's bodies to provide surgical access to surgical sites on the patient's body. Providing surgical access to surgical sites on a patient's body promotes favorable surgical conditions and increases the opportunity for successful results.

(3) Positioning the patient's body in one particular manner can provide a surgical team preferred and/or appropriate access to particular surgical sites, while other body positions may provide access to different surgical sites or different access to the same surgical site. As a surgical patient is often unconscious during a surgery, a surgical team may arrange a patient's body in various positions throughout the surgery. Surgical patient supports, such as operating tables, that accommodate a certain patient body position can provide surgical access to certain surgical sites while safely supporting the patient's body.

SUMMARY

(4) The present application discloses one or more of the features recited in the appended claims and/or the following features which, alone or in any combination, may comprise patentable subject matter:

(5) According to an aspect of the disclosure, a surgical patient support device may include a support frame having first and second support rails extending parallel to each other from a head end to a foot end of the patient support, a head-cross beam and a foot-cross beam connected to each of the support rails at the head end and foot end respectively, and a connection arm engaged with the head-cross beam, a platform mounted on the frame and including a torso section and a leg section, an actuator assembly coupled to the support frame and configured to support the leg section, and the leg section may be configured to move between a raised position and a lowered position.

(6) In some embodiments, the first and second support rails each may include a torso rail and a leg rail, the torso rails each extending from the head-cross beam towards the foot end to connect with the leg rail of the respective support rail, and each leg rail extending from connection with the torso rail of the respective support rail towards the foot end.

(7) In some embodiments, each leg rail may include a first sub-rail and a second sub-rail, and each first sub-rail may extend from connection with the torso rail of the respective support rail towards the foot end at an angle relative to the torso rail of the respective support rail.

(8) In some embodiments, each first sub-rail may extend from connection with the torso rail of the respective support rail towards the foot end at an angle of about 15 to about 35 degrees relative to the torso rail of the respective support rail.

(9) In some embodiments, each second sub-rail may extend from connection with the foot-cross beam for connection with the first sub-rail of the respective support rail. In some embodiments, in the lowered position the leg section of the platform may be parallel to each first sub-rail.

(10) In some embodiments, the actuator assembly may include at least one linear actuator configured for movement between a retracted position and an extended position to move the leg section of the support platform between the lowered position and the raised position.

(11) In some embodiments, the at least one actuator may include a cross link that extends between the leg rails of the support rails and a cross arm extending orthogonally from the cross link to support the at least one linear actuator. In some embodiments, the at least one linear actuator may be pivotably connected to the cross arm of the cross link.

(12) In some embodiments, each leg rail may include a jogged section that connects with the torso rail and a width defined between the leg rails of the support rails including the jogged section is wider than a width defined between the torso rails of the support rails.

(13) In some embodiments, the actuator assembly may be connected to the leg section of the platform on a bottom side thereof at a position spaced apart from the head end and the foot end.
(14) In some embodiments, the actuator assembly may include at least two actuators and a first of the at least two actuators is pivotably coupled to one of the support rails and a second of the at least two actuators is pivotably coupled to the other of the support rails, and each of the at least two actuators is pivotably coupled to the leg section of the platform and is configured for actuation to move the leg section of the support platform between the lowered and the raised positions.
(15) According to another aspect of the present disclosure, a surgical patient support system may include a base frame having a head elevator tower and a foot elevator tower each having a support bracket connected thereto and configured for translation of the support brackets between higher and lower positions; a support frame having first and second support rails extending parallel to each other from a head end to a foot end, a head-cross beam and a foot-cross beam extending between the first and second rails at the head end and foot end respectively; and connection arms including a head-connection arm engaged with the head-cross beam and coupled with the support bracket of the head tower and a leg-connection arm engaged with the leg-cross beam and coupled with the support bracket of the leg tower; a support platform coupled to the support frame and including a torso section and a leg section; an actuator assembly coupled to the support frame and configured to support the leg section; and the leg section is configured to move between a raised position and a lowered position to create leg break of a surgical patient in a lateral position.
(16) In some embodiments, the leg section of the support platform may be hingedly attached to the support frame to move between the raised position and the lowered position and the actuator assembly is pivotably connected to the leg section of the platform on a bottom side thereof.
(17) In some embodiments, the actuator assembly may be configured for operation between an extended and a retracted position and the extended position of the actuator assembly corresponds to the raised position of the leg section, and the retracted position of the at least one actuator corresponds to the lowered position of the leg section.
(18) In some embodiments, the lowered position may be arranged to contribute about 25° of leg break to a surgical patient in the lateral position. In some embodiments, the raised position may be arranged to contribute about 0° of leg break to a surgical patient in the lateral position. In some embodiments, the actuator assembly may include a linear actuator configured to rotate an axle.
(19) In some embodiments, the first and second rails may each include a torso rail which extends from the head end towards the foot end and the first and second rails define a constant width between the torso rails along the extension direction.
(20) According to another aspect of the present disclosure, a method of operating a surgical patient support may include transferring a patient onto the surgical patient support while maintaining a supine position, positioning the patient in a lateral position on the surgical patient support to permit access to the patient, operating the surgical patient support to provide leg break to the patient, and rotating the patient into a prone position while the surgical patient support remains rotationally fixed.
(21) In some embodiments, the method may include operating the surgical patient support to provide leg break to the patient includes lowering a leg section of a support platform of the surgical patient support to have an angle of between 0-35° with respect to a torso section of the support platform.
(22) According to another aspect of the disclosure, a surgical patient support extending from a head end to a foot end may include a support frame having first and second support rails extending parallel to each other between the head end and the foot end, a head-cross beam and a foot-cross beam connected to each of the support rails at the head end and foot end respectively, and a connection arm engaged with the head-cross beam, the first and second support rails each including a torso rail and a leg rail, the torso rails each extending from the head-cross beam towards the foot end to connect with the leg rail of the respective support rail, and each leg rail extends from

connection with the torso rail of the respective support rail towards the foot end, each leg rail includes a first sub-rail and a second sub-rail, and each first sub-rail extends from connection with the torso rail of the respective support rail towards the foot end at an angle relative to the torso rail of the respective support rail and each second sub-rail extends from connection with the foot-cross beam for connection with the first sub-rail of the respective support rail, a platform mounted on the support frame and including a torso section and a leg platform including a pivot end pivotably attached to the frame and a footward end proximate to the foot end of the patient support, the leg platform being configured to move between a raised position in which the leg platform is generally parallel with the torso platform and a lowered position in which the leg platform is pivoted out of parallel with the torso platform, an actuator assembly coupled to the support frame and configured to support the leg platform, and a protection sheath coupled to the second sub-rail of each of the leg rails to block against pinch point formation during movement of the leg platform.
(23) In some embodiments, the protection sheath may include a tray extending between opposite ends and an arm attached to each of the opposite ends of the tray. In some embodiments, the tray may be formed to have a shape that corresponds closely to the travel path of the leg platform between the raised and lowered positions to prevent pinch points.
(24) In some embodiments, the arms may each define an opening and a cavity extending from the opening into the respective arm, each arm being configured to receive one of the second sub-rails through the respective opening and into the respective cavity.
(25) In some embodiments, the tray may include an opening defined on a rear side thereof and a cavity extending from the opening into the tray for receiving the foot-cross beam therein.
(26) In some embodiments, the connection arm may extend through the opening in the tray. In some embodiments, the cavities of the arms may connect with the cavity of the tray.
(27) In some embodiments, each first sub-rail may extend from connection with the torso rail of the respective support rail towards the foot end at an angle of about 15 to about 35 degrees relative to the torso rail of the respective support rail. In some embodiments, in the lowered position the leg platform of the platform may be parallel to the first sub-rails.
(28) According to another aspect of the present disclosure, a surgical patient support may include a pair of elevator towers, a support frame extending between a head end and a foot end and coupled to one of the support towers at each end, the support frame including first and second support rails, a head-cross beam and a foot-cross beam connected to each of the support rails at the head end and foot end respectively, and a connection arm engaged with the head-cross beam, the first and second support rails each including a torso rail and a leg rail, the torso rails each extending from the head-cross beam towards the foot end to connect with the leg rail of the respective support rail, and each leg rail extends from connection with the torso rail of the respective support rail towards the foot end, each leg rail includes a first sub-rail and a second sub-rail, and each first sub-rail extends from connection with the torso rail of the respective support rail towards the foot end at an angle relative to the torso rail of the respective support rail and each second sub-rail extends from connection with the foot-cross beam for connection with the first sub-rail of the respective support rail, a platform mounted on the support frame and including a torso section and a leg section including a pivot end pivotably attached to the frame and a footward end proximate to the foot end of the patient support, the leg section being configured to move between a raised position in which the leg section is generally parallel with the torso section and a lowered position in which the leg section is pivoted out of parallel with the torso section, an actuator assembly coupled to the support frame and configured to support the leg section, and a protection sheath coupled to the second sub-rail of each of the leg rails to block against pinch point formation during movement of the leg section.
(29) In some embodiments, the protection sheath may include a tray extending between opposite ends and an arm attached to each of the opposite ends of the tray.
(30) In some embodiments, the tray may be formed to have a shape that corresponds closely to the travel path of the leg section between the raised and lowered positions to prevent pinch points.

(31) In some embodiments, the arms may each define an opening and a cavity extending from the opening into the respective arm, each arm being configured to receive one of the second sub-rails through the respective opening and into the respective cavity.
(32) In some embodiments, the tray may include an opening defined on a rear side thereof and a cavity extending from the opening into the tray for receiving the foot-cross beam therein.
(33) In some embodiments, the connection arm may extend through the opening in the tray. In some embodiments, the cavities of the arms may connect with the cavity of the tray.
(34) In some embodiments, each first sub-rail may extend from connection with the torso rail of the respective support rail towards the foot end at an angle of about 15 to about 35 degrees relative to the torso rail of the respective support rail. In some embodiments, in the lowered position the leg section of the platform may be parallel to each first sub-rail.
(35) These and other features of the present disclosure will become more apparent from the following description of the illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The detailed description particularly refers to the accompanying figures in which:
(2) FIG. **1** is a top perspective view of a surgical support including a patient support having a leg platform in a raised position;
(3) FIG. **2** is a top perspective view of the patient support of the surgical support as shown in FIG. **1** showing the leg platform in a lowered position;
(4) FIG. **3** is a bottom perspective view of the patient support of the surgical support as shown in FIG. **1**;
(5) FIG. **4** is a bottom perspective view of the patient support of the surgical support as shown in FIG. **2**;
(6) FIG. **5**A is a top perspective view of the patient support of the surgical support as shown in FIG. **1** showing that an actuator is extended to support the leg platform in the raised position;
(7) FIG. **5**B is an elevation view of the patient support of the surgical support as shown in FIG. **1** showing that in the raised position the patient's spine is generally aligned;
(8) FIG. **6**A is a top perspective view of the patient support of the surgical support as shown in FIG. **1** showing that the actuator is partly extended to support the leg platform in an intermediate position between raised and lowered positions;
(9) FIG. **6**B is an elevation view of the patient support of the surgical support as shown in FIG. **1** showing that in the intermediate position the patient's spine is slightly not aligned to create some leg break;
(10) FIG. **7**A is a top perspective view of the patient support of the surgical support as shown in FIG. **1** showing that the actuator is retracted to support the leg platform in lowered position;
(11) FIG. **7**B is an elevation view of the patient support of the surgical support as shown in FIG. **1** showing that in the lowered position the patient's spine is not aligned to create full leg break;
(12) FIG. **8** is a top perspective view of a patient support of another illustrative embodiment of the surgical support having a leg platform in a raised position;
(13) FIG. **9** is a top perspective view of the patient support as shown in FIG. **8** showing the leg platform in a lowered position;
(14) FIG. **10** is a top perspective view of a patient support of another illustrative embodiment of the surgical support having a leg platform in a raised position;
(15) FIG. **11** is a top perspective view of the patient support as shown in FIG. **10** showing the leg platform in a lowered position;
(16) FIG. **12** is a bottom perspective view of a patient support of another illustrative embodiment of

the surgical support having a leg platform in a raised position;
(17) FIG. **13** is a bottom perspective view of the patient support shown in FIG. **12** showing the leg platform in the lowered position;
(18) FIGS. **14**A-**14**F are pictorial flow sequence depictions of a support and a method of operating the surgical support for positioning a patient;
(19) FIG. **15** is an elevation view of the pictorial flow sequence portion depicted in FIG. **14**F showing the surgical support configured for accommodating a patient in a prone position and showing that an abdomen pad has been removed FIG. **16** is a perspective view of another surgical support that includes a patient support having a support frame supporting a platform that has a torso platform and a leg platform, the leg platform being pivotable between a raised position that is parallel with the torso platform and a lowered position that is inclined with respect to the torso platform, and showing that the surgical support includes a protection sheath coupled to the frame at the foot end of the surgical support to block against pinch points during movement of the leg platform between the raised and lowered positions;
(20) FIG. **17** is a perspective view of the patient support of the surgical support of FIG. **16** showing the leg platform in the lowered position and the protection sheath receiving upwardly extending rails of the support frame therein to couple the protection sheath with the frame and showing a horizontal beam of the frame received within the protection sheath;
(21) FIG. **18** is a perspective view of the patient support of FIG. **17** showing the leg platform in the raised position and the protective sheath including a tray and arms disposed on lateral sides of the tray, and showing the protective sheath having a shape that corresponds closely to the travel path of a foot end of the leg platform to block pinch points;
(22) FIG. **19** is a perspective front view of the protection sheath of FIGS. **16-18** showing the protection sheath having a curvature along a horizontal direction that corresponds closely to the shape of the foot end of the leg platform and showing the arms of the protection sheath defining cavities therein for receiving the rails of the frame; and
(23) FIG. **20** is a perspective rear view of the protective sheath of FIG. **19** showing the protective sheath including a cavity extending between the arms for receiving the beam of the frame and showing that the cavity for receiving the beam is in communication with the cavities of the arms that receive the rails.

DETAILED DESCRIPTION OF THE DRAWINGS

(24) For the purposes of promoting an understanding of the principles of the disclosure, reference will now be made to a number of illustrative embodiments illustrated in the drawings and specific language will be used to describe the same.
(25) Some surgical procedures, such as spinal fusion procedures, require particular access to various parts of a patient's spine. The course of a surgery can require a patient's body to be positioned for a period of time in several different manners, for example a lateral position for a lateral lumbar interbody fusion and a prone position for a posterior spinal fusion.
(26) For surgical procedures that are performed in the lateral body position (e.g., lateral lumbar interbody fusion), it can be desirable to articulate the patient's legs out of the sagittal plane along the coronal plane such that the patient's legs are generally out of parallel with the patient's torso, referred to as leg break. This leg break can provide appropriate access to certain surgical sites, for example certain lumbar areas. The present disclosure includes, among other things, surgical supports for accommodating various positions of a patient's body, including for example a lateral position with leg break and a prone position.
(27) In a first illustrative embodiment, a surgical support **10** includes a patient support **13** and a base **11** as shown in FIG. **1**. Base **11** supports patient support **13** above the floor to provide support to a surgical patient. Patient support **13** includes a frame **12**, a support platform **14**, and an actuator assembly **16**.
(28) As shown in FIG. **1**, frame **12** supports platform **14** that can support a patient, generally with

padding disposed between the patient and the platform **14** for comfort. The patient support **13** includes a head end **30**, a mid-section **32**, a foot end **34**, and left and right lateral sides **50**, **52**. Patient support **13** is configured to permit movement of the support platform **14** near the foot end **34** to provide leg break to a patient occupying the surgical support **10**.

(29) Base **11** includes elevator towers **19**, **21** as shown in FIG. **1**. Elevator towers **19**, **21** each include a bracket **17** and provide support to the frame **12** for vertical translation along the towers **19**, **21**. Bracket **17** of elevator tower **19** is connected to frame **12** of patient support **13** at head end **30**, and bracket **17** of elevator tower **21** is connected to frame **12** of the patient support **13** at foot end **34**.

(30) Frame **12** includes support rails **18**, **20** and first and second beams **22**, **24** as shown in FIG. **1**. Frame **12** is illustratively comprised of tubular members, but in some embodiments may include any one or more of solid, truss, and/or any combination of frame members. First beam **22** is illustratively arranged at the head end **30** and second beam **24** is arranged at the foot end **34** of the patient support **13**. Support rails **18**, **20** extend parallel to each other between beams **22**, **24** from the head end **30** to the foot end **34** of the patient support **13**.

(31) Support rail **18** illustratively connects with beam **22** on the left lateral side **50** (as depicted in FIG. **1**) of patient support **13** and extends footward to connect with beam **24** on the same lateral side **50** as shown in FIG. **1**. Support rail **20** illustratively connects with beam **22** on the right lateral side **52** (as depicted in FIG. **1**) of patient support **13** and extends footward to connect with beam **24** on the same lateral side **52** as shown in FIG. **1**. Frame **12** is configured to support the support platform **14**.

(32) Support platform **14** illustratively includes a torso platform **36** and a leg platform **38** as shown in FIG. **1**. Torso platform **36** extends from head end **30** to mid-section **32** of patient support **13**. Leg platform **38** extends from the mid-section **32** to the foot end **34** of the patient support **13**.

(33) Leg platform **38** is hingedly supported by frame **12** to pivot about an axis **25** extending laterally through surgical support **10** such that a footward end **42** of leg platform **38** is lowered relative to its headward end **40** to provide leg break to an occupying patient as shown in FIGS. **1-4**. Axis **25** is illustratively spaced apart from and perpendicular and/or orthogonal to axis **15**. In the illustrative embodiment as shown in FIGS. **1-4**, headward end **40** is hingedly connected to frame **12**, but footward end **42** of leg platform **38** is a free end having no direct connection with any support structure, for example, footward end **42** illustratively has no direct structural connection to frame **12**, bracket **17**, and/or tower **21**. In the illustrative embodiment as shown in FIGS. **3** and **4**, leg platform **38** includes hinged connections **63** each including a hinge block **65** and a hinge post **67**.

(34) Hinge blocks **65** are illustratively attached to a bottom side **71** of leg platform **38** at the headward end **40** thereof and in spaced apart relation to each other. One hinge post **67** illustratively extends from connection with one hinge block **65** in a direction away from the other hinge block **65** and parallel to the beams **22**, **24**. The other hinge post **67** illustratively extends from connection with the other hinge block **65** in a direction away from the one hinge block **65** and parallel to the beams **22**, **24**. One hinge post **67** is illustratively received in a bearing **69** of support rail **18** and the other hinge post **67** is illustratively received in a bearing **69** of support rail **20**, to permit pivotable movement of the leg platform **38**. In the illustratively embodiment, bearings **69** are embodied as plain bearings, but in some embodiments may include one or more of any suitable type of bearings, for example, roller bearings.

(35) Actuator assembly **16** assists in driving the leg platform **38** for pivoting movement between a raised position (shown in FIG. **1**) and a lowered position (shown in FIG. **2**). During pivoting of leg platform **38** by actuator assembly **16**, head platform **36** and all portions of frame **12** illustratively remain stationary.

(36) As shown in the illustrative embodiment of FIGS. **1-4**, support rails **18**, **20** of the frame **12** are disposed at respective left and right sides **50**, **52** of patient support **13** in spaced apart relation to

each other. Each support rail **18**, **20** includes a torso rail **54** and a leg rail **56**. Each torso rail **54** extends from the head end **30** to the mid-section **32** of the support device **10**.

(37) The torso rails **54** are each illustratively embodied as straight rails extending in parallel spaced apart relation to each other. The torso rails **54** are illustratively connected to opposite lateral ends of beam **22** as shown in FIG. **1**. Torso rails **54** on each lateral side **50**, **52** connect to one leg rail **56** on the corresponding lateral side **50**, **52** at the mid-section **32** of patient support **13**. In the illustrative embodiment, torso rails **54** are connected to their respective leg rails **56** by rigid connection such that rails **54**, **56** do not move relative to each other.

(38) Each leg rail **56** extends from the mid-section **32** to the foot end **34** of patient support **13** as shown in FIGS. **1** and **2**. Each leg rail **56** illustratively connects to one corresponding torso rail **56** at the mid-section **32** of patient support **13**. Each leg rail **56** includes a first sub-rail **58** and a second sub-rail **62** as shown in FIGS. **1** and **2**.

(39) In the illustrative embodiment, first sub-rail **58** of first rail **18** extends from mid-section **32** toward foot end **34** at angle α relative to its corresponding torso rail **54** of the same first rail **18**. In the illustrative embodiment, the first sub-rail **58** is straight and extends at angle α of about 25 degrees relative to its corresponding torso rail **54** of first rail **18**. In the illustrative embodiment, first sub-rail **58** of second rail **20** extends from the mid-section **32** toward the foot end **34** at angle α relative to the torso rail **54** of second rail **20**. In the illustrative embodiment, first sub-rail **58** of second rail **20** is straight and extends at angle α of about 25 degrees relative to its corresponding torso rail **54** of second rail **20**.

(40) As illustratively suggested in FIG. **1**, the angle α of each first sub-rail **58** is downward relative to their respective torso rails **54**, however, the indication of the relative direction downward is descriptive and is not intended to limit the orientation of the frame **12** of the support device **10**. In some embodiments, the first sub-rail **58** of each first and second rails **18**, **20** may have any angle relative to its corresponding torso rail **54** including but not limited to any angle within the range 0-40 degrees.

(41) Second sub-rails **62** are arranged in parallel spaced apart relation to each other as suggested in FIG. **1**. In the illustrative embodiment, second sub-rail **62** of first rail **18** is straight and is connected at its headward end **62***a* to a footward end **58***b* of the first sub-rail **58** of first rail **18** as shown in FIGS. **1** and **2**. Second sub-rail **62** of second rail **20** is straight and is connected at its headward end **62***a* to a footward end **58***b* of the first sub-rail **58** of second rail **20** as shown in FIGS. **1** and **2**. Second sub-rails **62** are connected on their footward ends **62***b* to opposite ends of beam **24**.

(42) In the illustrative embodiments shown in FIGS. **1** and **2**, first and second sub-rails **58**, **62** of the same one of first and second rails **18**, **20** are embodied as each being welded to each other and also to a reinforcement plate **70**. In some embodiments, first and second sub-rails **58**, **62** of the same one of first and second rails **18**, **20** are connected to each other and/or to plate **70** by one or more of welding, brazing, integral formation, pinning, bolting, and/or any other suitable manner of joining. In some embodiments, additional sub-rails connect the first sub-rail **58** to the second sub-rail **62** for the same first and second rail **18**, **20**, for example, a third sub-rail may connect to the footward end **58***b* of the first sub-rail of one of the first and second rails **18** and the headward end **62***a* of the second sub-rail **62** of the same one of the first rail and second rail **18**.

(43) In the illustrative embodiment as shown in FIGS. **1**, **3**, and **4**, actuator assembly **16** is connected between frame **12** and platform **14** to provide movement and positioning of platform **14** relative to the torso platform **36**. As shown in FIG. **3**, actuator assembly **16** illustratively includes an actuator **68**, a cross link **64**, and a cross arm **66**. Cross link **64** connects to frame **12**.

(44) Cross link **64** includes a first end **64***a* and a second end **64***b* as shown in FIG. **3**. Cross link **64** illustratively connects at its first end **64***a* to first support rail **18** and extends to a second end **64***b* that connects to second support rail **20**. Cross link **64** is illustratively embodied as arranged parallel to beams **22**, **24** and connecting on either end **64***a*, **64***b* to the first sub-rails **58**. In some embodiments, cross link **64** may connect to any portion of the frame **12** suitable to provide support

to actuator **68**. Cross link **64** supports cross arm **66**.
(45) Cross arm **66** illustratively connects to the cross link **64** as shown in FIGS. **1-4**. Cross arm **66** illustratively connects to cross link **64** about midway between lateral sides **50**, **52** of patient support **13** and extends from cross link **64** in a direction generally away from the platform **14** to support actuator **68**. In the illustrative embodiment, cross arm **66** comprises two plates each connected to cross link **64** at one end and connected at their other end by pinned connection to actuator **68**. In some embodiments, cross link **64** and/or cross arm **66** may include one or more of a tubular member, solid member, truss member, and/or any combination thereof to support actuator **68** for moving the leg platform **38** between the raised and lowered positions.
(46) Actuator **68** illustratively includes a first end **68***a* pivotably connected to the cross arm **66** and a second end **38***b* pivotably connected to leg platform **38** as shown in FIGS. **3** and **4**. In the illustrative embodiment, actuator **68** is pivotably attached to a bottom side **71** of leg platform **38** by a pinned connection. Actuator **68** is illustratively embodied as a linear actuator configured to move between retracted (FIG. **4**) and extended (FIG. **3**) positions. Actuator **68** is illustratively embodied as an electro-mechanical actuator powered by an electric motor, for example, a suitable actuator is Actuator LA23 available from LINAK U.S. Inc. of Louisville, Kentucky.
(47) In some embodiments, actuator **68** may include one or more of a mechanical, hydraulic, pneumatic, any/or any other type of actuator suitable for assisting movement of the leg platform **38** between raised and lowered positions. In some embodiments, actuator **68** may be attached by one or more of a hinge, ball joint, and/or any type of connection to provide support to actuator **68** for moving the leg platform **38** between the raised and lowered positions. Actuator **68** is configured to drive the leg platform **38** for pivoting movement between the raised (FIG. **4**) and lowered (FIG. **3**) positions to create leg break to a patient occupying patient support **13**.
(48) As shown in FIGS. **5**A-**7**B, actuator **68** is illustratively configured to operate between extended and retracted positions to pivotably move leg platform **38** between raised and lowered positions to create leg break to a patient occupying patient support **13**. As shown in FIGS. **5**A and **5**B, leg platform **38** is arranged in the raised positioned when actuator **68** is in the extended positioned. In the illustrative embodiment as shown in FIGS. **5**A and **5**B, in the raised position, leg platform **38** is arranged generally coplanar with torso platform **36**. In some embodiments, the raised position of leg platform **38** may include a slight angle with respect to torso platform, for example, an angle in the range of about −5 to about 5 degrees. In the illustrative embodiment as shown in FIG. **5**B, in the raised position of the leg platform **38**, the patient's spine in generally aligned and creates little or no leg break.
(49) As shown in FIGS. **6**A and **6**B, the leg platform **38** is arranged in an intermediate position which is defined between the lowered and raised positions. The leg platform **38** is arranged in the intermediate position when actuator **68** to is in an intermediate extension position which is defined between the retracted and extended positions of actuator **68**. In the illustrative embodiment, in the intermediate position of the leg platform **38** as shown in FIGS. **6**A and **6**B, the leg platform **38** is generally arranged at an angle α′, between about 0 and about 25 degrees, relative to the torso platform **36**. In some embodiments, in the intermediate position, the leg platform **28** may be arranged at any angle α′, between about −5 and about 40 degrees, relative to the torso platform **36**. In the illustrative embodiment as shown in FIG. **6**B, in the intermediate position of leg platform **28**, the patient's spine is flexed, i.e., slightly not aligned, to create some leg break.
(50) As shown in FIGS. **7**A and **7**B, the leg platform **38** is arranged in the lowered position when actuator **68** is in the retracted position. In the illustrative embodiment, in the lowered position of the leg platform **38** as shown in FIGS. **7**A and **7**B, the leg platform **38** is generally arranged at an angle α equal to about 25 degrees, relative to the torso platform **36**. In some embodiments, in the lowered position, the leg platform **38** may be arranged at any angle α from about 0 to about 40 degrees, relative to the torso platform **36**. In the illustrative embodiment as shown in FIG. **7**B, in the lowered position of leg platform **28**, the patient's spine is not aligned, for example, greatly not

aligned, to create full leg break.

(51) Beams **22**, **24** each couple to a floating arm **44** that is configured for connection to support towers **19**, **21** via brackets **17** as shown in FIGS. **1-4**. Each floating arm **44** is illustratively movably connected to its respective beam **22**, **24** for pivoting movement to accommodate rotation of patient support **13** about axis **15** under configuration of frame **12** with different vertical positions of its head end **30** and foot end **34** without binding, although the present disclosure does not require rotation of the patient support **13**.

(52) Each floating arm **44** includes a connection tube **46**. Connection tube **46** is connected to its floating arm **44** as shown in FIGS. **2-4**. In the illustrative embodiment, connection tube **46** is a hollow cylinder connected at an intermediate point along its length to the floating arm **44** and configured to receive connection pin **48** therethrough to pin the floating arm **44** to bracket **17** of one of the elevator towers **19**, **21** as suggested in FIG. **1**. In some embodiments, the connection between frame **12** and bracket **17** may be configured similar to the motion coupler and its related components disclosed in U.S. Patent Application Publication No. 2013/0269710 by Hight et al., for example in FIGS. 41-44 and 69-73, and the contents of U.S. Patent Application Publication No. 2013/0269710 are hereby incorporated by reference including both the particulars of the motion coupler and its related components and the remainder of the disclosure in its entirety.

(53) Referring now to a second illustrative embodiment shown in FIGS. **8** and **9**, a patient support **213** includes a frame **212**, a platform **214**, and an actuator assembly **216**. Patient support **213** is configured for use in surgical support **10** and is similar in many respects to the patient support **13** shown in FIGS. **1-7** and described herein. Accordingly, similar reference numbers in the 200 series indicate features that are common between patient support **213** and patient support **13** unless indicated otherwise. The description of patient support **13** is equally applicable to patient support **213** except in instances when it conflicts with the specific description and drawings of patient support **213**.

(54) Frame **212** includes support rails **218**, **220** and first and second beams **222**, **224**. Support rails **218**, **220** extend parallel to each other between beams **222**, **224** from the head end **30** to the foot end **34** of patient support **213**.

(55) Support rail **218** illustratively connects with beam **222** on the left lateral side **50** (as depicted in FIG. **8**) of patient support **213** and extends footward to connect with beam **224** on the same lateral side **50** as shown in FIG. **8**. Support rail **220** illustratively connects with beam **222** on the right lateral side **52** (as depicted in FIG. **8**) of patient support **213** and extends footward to connect with beam **224** on the same lateral side **52** as shown in FIG. **8**. Frame **212** is configured to support the support platform **214**.

(56) Support platform **214** illustratively includes a torso platform **236** and a leg platform **238** each having supporting padding **286** as shown in FIG. **8**. Leg platform **238** is hingedly supported by frame **212** to pivot such that a footward end **242** of leg platform **238** is lowered relative to its headward end **240** to provide leg break to an occupying patient. Actuator assembly **216** assists in driving the leg platform **238** for pivoting movement between a raised position (shown in FIG. **8**) and a lowered position (shown in FIG. **9**). In the illustrative embodiment as shown in FIGS. **8** and **9**, headward end **240** is hingedly connected to frame **212**, but footward end **242** of leg platform **238** is a free end having no direct connection with any support structure, for example, footward end **242** illustratively has no direct structural connection to frame **212**, bracket **17**, and/or tower **21**.

(57) In the illustrative embodiment as shown in FIGS. **8** and **9**, support rails **218**, **220** of the frame **212** are disposed at respective left and right sides **50**, **52** of patient support **213** in spaced apart relation to each other. Each support rail **218**, **220** includes a torso rail **254** and a leg rail **256**. Each torso rail **254** extends from the head end **30** to the mid-section **32** of patient support **13** to connect with its respective leg rail **256**. In the illustrative embodiment, torso rails **254** are connected to their respective leg rails **256** by rigid connection such that rails **254**, **256** do not move relative to each other.

(58) Each leg rail **256** extends between the mid-section **32** to the foot end **34** of the patient support **213** as shown in FIGS. **8** and **9**. Each leg rail **256** illustratively connects to a corresponding torso rail **256** at the mid-section **32** of the patient support **213**. Each leg rail **256** includes a first sub-rail **258** and a second sub-rail **262** as shown in FIG. **8**.
(59) In the illustrative embodiment, each first sub-rail **258** of each support rail **218**, **220** includes a first segment **258***a* and a second segment **258***b* as shown in FIG. **9**. First segment **258***a* of each rail **258** illustratively extends from mid-section **32** towards foot end **34** at angle β relative to its corresponding torso rail **254** of the same rail **218**, **220**. First segment **258***a* connects to and is illustratively integral with second segment **258***b.*
(60) Second segment **258***b* extends from first segment **258***a* towards the foot end **34** as shown in FIG. **9**. In the illustrative embodiment, second segment **258***b* of first sub-rail **258** is straight and extends from first segment **258***a* parallel to its corresponding torso rail **254**. Second segment **258***b* illustratively connects to second sub-rail **262**.
(61) As illustratively suggested in FIG. **8**, the angle β of each first segment **258***a* is about 30 degrees. In some embodiments, first segment **258***a* of first sub-rail **258** of support rails **218**, **220** may have any angle relative to its corresponding torso rail **254** including but not limited to any angle within the range 0-40 degrees.
(62) Second sub-rails **262** are arranged in parallel spaced apart relation to each other as suggested in FIGS. **8** and **9**. In the illustrative embodiment, second sub-rails **262** connect to their respective first sub-rails **256** and extend perpendicularly therefrom as shown in FIGS. **8** and **9**. Second sub-rails **262** each connect to opposite lateral ends of second beam **224**.
(63) Actuator assembly **216** includes actuators **268** as shown in FIG. **8**. Each actuator **268** has first end **268***a* pivotably coupled to frame **212** and second end **268***b* pivotably coupled to support platform **214** as shown in FIGS. **8** and **9**. Illustratively, first end **268***a* of one of the actuators **268** is coupled to leg rail **256** of one of the support rails **218**, **220**, and first end **268***a* of the other actuator **268** is illustratively coupled to leg rail **256** of the other support rail **218**, **220**. Ends **268***a* of each actuator **268** are illustratively connected to frame **212** by brackets **269**. Illustratively, second end **268***b* of one of the actuators **268** is coupled to a bottom side of the leg platform **238**, and second end **268***b* of the other actuator **268** is illustratively coupled to the bottom side of the leg platform **238** in spaced apart relation to the second end **268***b* of the one actuator **268**.
(64) Leg platform **238** is illustratively includes tapered sections **253** located at the footward end **242** as shown in FIGS. **10** and **11**. Tapered sections **253** are illustratively defined by chamfers of the leg platform **238**. Each tapered section **253** includes a channel **255** defined in a bottom surface of leg platform **238**. Channels **255** are configured to accommodate actuators **268** therein when the leg platform **238** is in the lowered position as shown in FIG. **11**.
(65) Referring now to a third illustrative embodiment shown in FIGS. **10** and **11**, a patient support **313** includes frame **312**, a platform **314**, and an actuator assembly **316**. Patient support **313** is configured for use in surgical support **10** and is similar in many respects to the patient supports **13**, **213** shown in FIGS. **1-9**B and described herein. Accordingly, similar reference numbers in the 300 series indicate features that are common between patient support **313** and any of patient supports **13**, **213** unless indicated otherwise. The description of patient supports **13**, **213** is equally applicable to patient support **313** except in instances when it conflicts with the specific description and drawings of patient support **313**.
(66) Frame **312** includes support rails **318**, **320** and first and second beams **322**, **324**. Support rails **318**, **320** extend in spaced apart relation to each other between beams **322**, **324** from the head end **30** to the foot end **34** of the patient support **310**.
(67) Support rail **318** illustratively connects with beam **322** on the left lateral side **50** (as depicted in FIG. **10**) of patient support **313** and extends footward to connect with beam **324** on the same left lateral side **50** as shown in FIG. **10**. Support rail **320** illustratively connects with beam **322** on the right lateral side **52** (as depicted in FIG. **10**) of patient support **313** and extends footward to connect

with beam **324** on the same right lateral side **52** as shown in FIG. **10**.

(68) Support rails **318**, **320** each include a torso rail **354** and a leg rail **356** as shown in FIGS. **10** and **11**. Each torso rail **354** extends from the head end **30** to the mid-section **32** of the patient support **313**. Torso rails **354** are each illustratively embodied as straight rails extending in parallel spaced apart relation to each other. Torso rails **354** are illustratively connected to beam **322** at opposite lateral ends thereof as shown in FIG. **10**. Torso rails **354** on each lateral side **50**, **52** connect to one leg rail **356** on the corresponding lateral side **50**, **52** at the mid-section **32** of the patient support **313**.

(69) Each leg rail **356** extends from the mid-section **32** to the foot end **34** of patient support **13** as shown in FIGS. **10** and **11**. Each leg rail **356** illustratively connects to one corresponding torso rail **354** at the mid-section **32** of patient support **313**. At the mid-section **32**, the leg rails **356** are in spaced apart relation to each other defining a first distance w illustratively equal to a distance between rails **354**. Each leg rail **356** is formed to include a jog **357**.

(70) Each jog **357** is a bent section of its leg rail **356** as shown in FIGS. **10** and **11**. Each jog **357** illustratively includes a section of one leg rail **356** which is bent outwardly in a direction away from the other leg rail **356** such that the leg rails **356** are in spaced apart relation to each other defining a second distance W greater than the first distance w defined between rails **354**. In the illustrative embodiment, jog **357** of each leg rail **356** extends outwardly away from the other leg rail **356** by an equal amount. Leg rails **356** along their entire length are illustratively coplanar with the torso rails **354**. Jogs **357** are illustratively embodied as integral sections of rails **318**, **320** that are curved as a part of formation, but in some embodiments may include distinct rail portions joined by any suitable joining manner, for example, fastening and/or welding.

(71) Support platform **314** illustratively includes a torso platform **336** and a leg platform **338** as shown in FIG. **10**. Leg platform **338** is hingedly supported by frame **312** to pivot such that a footward end **342** of leg platform **338** is lowered relative to its headward end **340** to provide leg break to an occupying patient as shown in FIGS. **10** and **11**. Leg platform **338** is arranged between leg rails **356** and is configured for movement between the leg rails **356**. Actuator assembly **316** assists in driving the leg platform **338** for pivoting movement between a raised position (shown in FIG. **10**) and a lowered position (shown in FIG. **11**). In the illustrative embodiment as shown in FIGS. **10** and **11**, headward end **340** is hingedly connected to frame **12**, but footward end **342** of leg platform **338** is a free end having no direct connection with any support structure, for example, footward end **342** illustratively has no direct structural connection to frame **312**, bracket **17**, and/or tower **21**.

(72) In the illustrative embodiment shown in FIG. **10**, actuator assembly **316** includes gas spring actuators **368** configured to assist manual operation of leg platform **338** between raised and lowered positions. Bracket **329** connected to the underside of leg platform **338** and has a U-shaped portion in which the leg rails **356** rest when leg platform **338** is in the raised position as shown in FIG. **10**. In the illustrative embodiment, an end of one actuator **368** is pivotably attached to an outer lateral end of beam **324**, and another end of the same actuator **368** is pivotably attached to an actuator bracket **369**. An end of the other actuator **368** is pivotably attached to another outer lateral end of beam **324**, and another end of the same actuator **368** is pivotably attached to an actuator bracket **369**. Actuator brackets **369** are illustratively connected to leg platform **338** at opposite lateral sides **50**, **52** to provide pivotable operation assistance thereto. In some embodiments, such as the embodiment as shown in FIG. **11**, actuators **368** are configured for full powered actuation independent of manual operation, for example, configuration to drive the full load of leg platform **338** and an occupying patient and/or including connection to a control system for activation of the actuators **368**. In some embodiments, actuators **368** may be omitted in favor of a fully manual operation of leg platform **338**.

(73) Regardless of whether actuators are gas springs or powered linear actuators, the positing of leg platform **228** in the raised and lowered positions is generally as depicted in FIGS. **10** and **11**. The

gas springs contemplated are locking gas springs that are released via actuation of a release handle as is well known in the art. Such a release handle may be located in the vicinity of the bracket **369**, for example. Actuation of the release handle adjacent either bracket **369** releases both gas springs via suitable cabling and/or linkages. In the case of linear actuators, an electrical cable from actuators **368** plugs into a port of base **11** so that an electrical control panel of base **11** is used to control operation of the actuators **368**.
(74) Torso platform **336** comprises head platform **336***a*, a chest platform **336***b*, a hip platform **336***c*, and arm platforms **337** as shown in FIGS. **10** and **11**. In the illustrative embodiment, each of head platform **336***a*, chest platform **336***b*, hip platform **336***c*, and arm platforms **337** comprise body-part specific supports and padding that are independently attached to the frame **312** and configured to provide a comfortable interface to the specific parts of the patient's body in a variety of positions. In the illustrative embodiment shown in FIGS. **10** and **11**, hip platform **336***c* illustratively includes two hip pads that are selectively configurable in either of a flat position (FIG. **11**) to accommodate supine and/or lateral positioning, or an angled position (FIG. **10**) to accommodate prone positioning.
(75) Chest platform **336***b* includes breast platform **339** and abdomen platform **341** as shown in FIG. **10**. In the illustrative embodiment, breast platform **339** has a U-shape. Breast platform **339** is configured to support a patient's upper chest, but not her abdomen while the patient is in the prone position. Breast platform **339** illustratively surrounds abdomen platform **341** on three sides thereof.
(76) Abdomen platform **341** is arranged between chest platform **339** and hip platform **336***c* as shown in FIG. **10**. As shown in FIG. **10**, abdomen platform **341** is arranged in a raised position generally coplanar with chest platform **339** to support the patient's middle body in certain positions, for example, the lateral and supine positions. As described herein with respect to abdomen pad **1300** shown in FIG. **15**, abdomen platform **341** is configurable into a lowered position to allow the abdomen of a patient in the prone position to hang downwardly and/or sag relative to the torso platform **336** of patient support **313**. Allowing the patient's abdomen to sag can provide particular spine arrangement while the patient is lying in the prone position.
(77) Referring now to a fourth illustrative embodiment shown in FIGS. **12** and **13**, a patient support **413** includes a frame **412**, a platform **414**, and an actuator assembly **416**. Patient support **413** is configured for use in in surgical support **10** and is similar in many respects to patient supports **13**, **213**, **313** shown in FIGS. **1-11** and described herein. Accordingly, similar reference numbers in the 400 series indicate features that are common between patient support **413** and any of patient supports **13**, **213**, **313** unless indicated otherwise. The description of patient supports **13**, **213**, **313** is equally applicable to patient support **413** except in instances when it conflicts with the specific description and drawings of patient support **413**.
(78) Actuator assembly **416** is configured to operate to drive a leg platform **438** between raised (FIG. **12**) and lowered (FIG. **13**) positions. Actuator assembly **416** includes an actuator **468**, a lever **472**, an axle **474**, a transmission bar **478**, a slider **480**, and a slider rail **484**. Actuator **468** illustratively applies force to lever **472** to rotate axle **474** and transmission bar **478**, such that slider **480** moves along slider rail **484** to move the leg platform **438** between raised and lowered positions as suggested in FIGS. **12** and **13**.
(79) Actuator **468** has an end **468***a* pivotably coupled to a bottom side **471** of leg platform **438** and another end **468***b* pivotably coupled to lever **472**. In the illustrative embodiment, actuator **468** is a linear actuator configured to operate between extended (FIG. **12**) and retracted positions (FIG. **13**). Lever **472** is illustratively configured to rotate to transfer linear movement of actuator **468** to pivoting movement of axle **474** to drive leg platform **438** between raised and lowered positions.
(80) Lever **472** is pivotably attached to end **468***b* of actuator **468** as shown in FIG. **13**. Lever **472** is connected to and fixed against rotation with respect to axle **474**. Axle **474** is rotatably supported by leg platform **438**. Axle **474** includes first and second ends **474***a*, **474***b*. Each end **474***a*, **474***b* is illustratively supported for rotation at by a mount **476** that extends perpendicularly from bottom

side **471** of leg platform **438**. Axle **47** is illustratively fixed against rotation with respect to transmission bar **478**.

(81) Transmission bar **478** is configured to transmit rotational force from axle **474** to frame **412** to drive the leg platform **438** between lowered and raised positions as shown in FIGS. **12** and **13**. Transmission bar **478** is illustratively connected to end **474***b* of axle **474**. Transmission bar extends from the axle **474** to pivotably connect with a slider **480**. Slider **480** is configured to be mounted onto a slider rail **484** to drive leg platform **438** between raised and lowered positions.

(82) Slider rail **484** is mounted to frame **412** as shown in FIGS. **12** and **13**. Slider rail **484** is illustratively attached to support rail **418** below the support rail **418** and extends parallel thereto. Slider rail **484** including a headward end **484***a* and footward end **484***b* each connected to support rail **418** by rail mounts **486** such that slider rail **484** is in spaced apart relation to support rail **418**. Movement of slider **480** along the slider rail **484** corresponds to the position of leg platform **438** between the raised and lowered positions. In some embodiments, two bars **479**, sliders **480**, and rails **482** are provided at opposite sides of patient support **413** and both operate as just described.

(83) According to another aspect of the disclosure, a surgical support and method of operating the surgical support are shown in FIGS. **14**A-**14**F. During a surgery, it may be desirable to place the patient in a first position, for example a lateral position, for a period of time and then to reposition the patient in a second position, for example a prone position. A surgical support **1000** is configured to accommodate both lateral and prone positions of the patient. Surgical support **1000** includes patient support **1013**.

(84) Surgical support **1000** is substantially similar to surgical support **10**, and patient support **1013** is substantially similar to patient support **413** shown in FIGS. **12** and **13** and described herein. Accordingly, similar reference numbers in the 1000 series indicate features that are common between patient support **1013** and patient support **413** unless indicated otherwise. The description of patient supports **413** is equally applicable to patient support **1013** except in instances when it conflicts with the specific description and drawings of patient support **1013**.

(85) A patient is positioned in proximity to surgical support **1000** on a support surface of a transport device such as a stretcher as shown in FIG. **14**A. The patient is typically transported while lying in the supine position. The patient is transferred to surgical support **1000** in the supine position as shown in FIG. **14**B.

(86) During a surgical procedure, the surgical team moves the patient's body into the lateral position as shown in FIG. **14**C. This involves rotating the patient by about 90 degrees onto the patient's side without rotating the patient support **1000** relative to base **11**. In the illustrative embodiment, the lateral position affords access to certain surgical sites on the patient's body, for example the spine. In the illustrative embodiment as shown in FIG. **14**C, various limb supports **1100** are selectively attached to frame **1012** and/or positioning devices **1200** are placed in contact with the patient to finely adjust the patient's body for surgical access. Positioning device **1200** is illustratively embodied as a surgical pillow but may include any of clamps, straps, cushions, bladders, and/or supports.

(87) Surgical support **1000** is operated to lower leg platform **1038** relative to torso platform **1036** to provide leg break to the patient as shown in FIG. **14**D. Leg portion **1038** is operated to achieve a desired position between the raised and lowered positions to produce the desired amount of leg break, illustratively the lowered position as shown in FIG. **14**D. Leg break provides access to certain surgical sites during certain portions of surgical procedures, for example, to spinal areas during a lateral spinal fusion, more specifically a lateral lumbar interbody fusion.

(88) Surgical support **1000** is operated to remove leg break from the patient as shown in FIG. **14**E. Leg portion **1038** is operated to achieve the raised position. Limb supports **1100** and positioning devices **1200** are illustratively removed and replaced with limb supports **1101** and positioning devices **1201** for supporting the patient while lying in the prone position.

(89) The surgical team moves the patient's body into the prone position as shown in FIG. **14**F. This

illustratively involves rotating the patient by about 90 degrees onto the patient's front without rotating the patient support **1000** relative to base **11**. The prone position provides access to certain surgical sites to permit certain surgical procedures, for example, posterior spinal fusion.
(90) An abdomen platform **1300** is illustratively pivoted downwardly away from the patient's body to accommodate the patient's body in the prone position as shown in FIG. **15**. The abdomen platform **1300** is configured to attach to a frame **1012** to be selectively positioned between a raised position suggested in FIGS. **14**A-**14**F to support the patient, and a lowered position as shown in FIG. **15** to permit the patient's abdomen to hang downwardly relative to torso platform **1036**. Lowering of the abdomen platform **1300** can enhance the positioning of the spine the patient's spine in position for surgery.
(91) The surgical support **1000** accommodates various patient body positions including lateral position with leg break and prone position. The surgical support **1000** thus provides access to surgical sites of the patient's body in various body positions without the need to rotate surgical support **1000** relative to base **11**.
(92) The present disclosure includes, among other things, the notion that during spinal surgery, the surgeon often needs to “break” the patient's legs. This means they are bent down below the horizontal plane of their torso in order to open the lateral disk space in their spine. Various supports are disclosed herein that can allow a surgeon to drop the patient's legs. This can be accomplished through one or more of a passive/manual joint, electric actuator(s), and/or pneumatic actuator(s). The leg drop section allows a surgeon to position the patient's legs in a range of angular positions, such as from 0 to 30 degrees.
(93) Clinically, this allows a surgeon to increase the vertebral spacing of the lumbar spine to gain access to the necessary disk space. This can be done before and/or during surgery. The device can have a major structural frame spanning two columns of the table. Within this frame, there is a secondary rotatable structure that allows the patient's legs to drop in between the structural frame or relative to the structural frame, depending upon the embodiment. In one aspect, the angle is manually adjusted and then locked at the desired position. In another aspect, a spring force, such as that provided by gas springs is applied to aid in supporting the patient's legs. In another aspect, an electric or pneumatic actuator drives the leg platform or section to the desired position. A leg drop section allows the surgeon to use the same table for lateral and prone surgeries. As the lateral surgery is often followed up immediately on the same patient with a prone surgery, this eliminates the need of transferring the patient to a separate table or rotating the patient to a different table top structure that attached to base **11**. The disclosed devices have additional clearance for imaging equipment (such as a C Arm) and is desirable for spinal surgeries.
(94) The present disclosure includes, among other things, a discussion of supports that allows a surgeon to complete a lateral lumbar interbody fusion with posterior fusion on one support frame. Such devices may allow a patient to be transferred from a stretcher onto the device in supine position, the patient to be rotated into a lateral position using a drawsheet, and/or the patient to be rotated into a prone position using a drawsheet. Patient support pads of the device can be adjustable and/or adaptable to all three positions eliminating the needs to transfer the patient onto an additional device during the procedure. The device may include dual parallel carbon fiber rails that can accommodate various pad attachments.
(95) The support pads may lay flat to accommodate a supine and lateral patient. When the patient is in the prone position, the hip pads can be adjusted so that they are angled to properly support the patient's hips and the pad underneath the patient abdomen may drop away so that the abdomen can hang free. The leg support sections disclosed herein are hinged near the hip of the patient so that the legs can be dropped below horizontal in the lateral position as well as in the prone position. The disclosed devices may eliminate the need to transfer the patient to an additional device during lateral to prone procedure, eliminate the need to log-roll a patient from the stretcher into the prone position 180 degrees, clear access to surgical sites by eliminating vertical supports, provide a

support top that does not need to rotate because the patient is rotating on top of the support platform, provide that the patients legs can be dropped in lateral as well as prone positions because of the breaking support platform.
(96) The present disclosure includes, among other things, a discussion of rigid lateral patient support frames that can flex the patient at the hip by a hinged support section. Utilizing a linkage and actuator, the patient's legs can be safely raised and lowered with a single low powered actuator, reducing complexity and other aspects of a two actuator design. The device may consist of a carbon fiber frame lateral leg support section that is mounted to by hinge to a main support frame. A linear actuator can be mounted to an underside of the leg drop section on one end and then connected to a moment arm on the other end. The moment arm may be directly connected to a rotary shaft. Attached to each end of the rotary shaft may be another linkage that transmits the power of the actuator to a linear rail. As the actuator pushes or pulls, the linkage can be forced to slide along the rail which raises and lowers the leg section. Such an arrangement may allow for a patient to be flexed in a lateral position, for the support top to be cheap, light, and easy to connect to the existing product bases, and/or for a single actuator to be used in lieu of two actuators.
(97) According to another aspect of the present disclosure, a surgical support **2000** and method of operating the surgical support **2000** are shown in FIGS. **16-20**. During a surgery, it may be desirable to place the patient in a first position, for example a lateral position, for a period of time and then to reposition the patient in a second position, for example a prone position. Surgical support **2000** is configured to accommodate both lateral and prone positions of the patient. Surgical support **2000** includes a first patient support **2012** configured to support the patient in the supine and lateral positions during surgery and a second patient support **2013** configured to support the patient in the prone position during surgery. Supports **2012**, **2013** are oriented at about 90° with respect to each other. Thus, supports **2012**, **2013** are rotated during surgery so that one or the other of supports **2023**, **2013** underlies and supports the patient.
(98) Surgical support **2000** is substantially similar to surgical support **10** as described above. Accordingly, the description and illustrations of surgical support **10** is equally applicable to surgical support **2000** except in instances of conflict with the specific description and drawings of surgical support **2000**.
(99) Surgical support **2000** includes a base **2011** as shown in FIG. **16**. Base **2011** supports patient supports **2012**, **2013** above the floor to provide support to the surgical patient. Patient support **2012** includes a frame **2015**, a support platform **2014** having support padding **2286** disposed thereon, and an actuator assembly **2016**. The support platform **2014** is operable to provide leg break to a patient occupying the surgical support **2000** while lying in the lateral position.
(100) As shown in FIG. **16**, frame **2015** supports the support platform **2014** that, in turn, supports the patient, generally with padding disposed between the patient and the support platform **2014** for comfort. Each of the patient supports **2012**, **2013** includes a head end **30**, a mid-section **32**, a foot end **34**, and right and left lateral sides **50**, **52**. Patient support **2012** is configured for leg break action of the support platform **2014** that includes movement of a leg platform **2038** between a raised position in which leg platform **2038** is generally parallel with a torso platform **2036** of support **2012** (as shown in FIG. **18**) and a lowered position in which the leg platform **2038** is pivoted out of parallel to an inclined position with respect to the torso platform **2036** (as shown in FIG. **17**) to provide leg break to the patient occupying the surgical support **2000**. Patient support **2012** illustratively includes a protection sheath **2070** coupled to the frame **2015** proximate to the foot end **34** to provide pinch protection while operating the leg platform **2038** for movement.
(101) Base **2011** includes elevator towers **19**, **21** as shown in FIG. **16**. Elevator towers **19**, **21** each carry a support bracket **2017** to provide support to the patient support **2012** for vertical raising, lowering, and tilting when one or both of the towers **19**, **21** are operated to extend or retract. One portion of support bracket **2017** of elevator tower **19** is connected to frame **2015** of patient support **2012** at the head end **30**, and one portion of bracket **2017** of elevator tower **21** is connected to

frame **2015** of the patient support **2012** at the foot end **34**. Another portion of support bracket **2017** of elevator tower **19** is connected to patient support **2013** at the head end **30**, and another portion of bracket **2017** of elevator tower **21** is connected to the patient support **2013** at the foot end **34**.

(102) Frame **2015** includes support rails **2018**, **2020** and first and second beams **2022**, **2024** as shown in FIG. **16**. Rails **2018**, **2020** extend generally in the longitudinal dimension of surgical support **2000** and beams **2022**, **2024** extend generally horizontally in the lateral dimension of surgical support **2000** when patient support **2012** is supported in orientation shown in FIGS. **16-18**. Frame **2015** is illustratively comprised of tubular members, but in some embodiments may include any one or more of solid, truss, and/or any combination of frame members. In some embodiments, rails **2018**, **2020** and beams **2022**, **2024** are made primarily of radiolucent materials such as carbon fiber materials. First beam **2022** is illustratively arranged at the head end **30** and second beam **2024** is arranged at the foot end **34** of the patient support **2012**. Support rails **2018**, **2020** extend parallel to each other between beams **2022**, **2024** from the head end **30** to the foot end **34** of the patient support **2012**.

(103) Support rail **2018** illustratively connects with beam **2022** on the right lateral side **50** (as depicted in FIG. **16**) of patient support **2013** and extends footwardly to connect with beam **2024** on the same lateral side **50** as shown in FIG. **16**. Support rail **2020** illustratively connects with beam **2022** on the left lateral side **52** (as depicted in FIG. **16**) of patient support **2013** and extends footwardly to connect with beam **2024** on the same lateral side **52** as shown in FIG. **16**. Frame **2015** is configured to support the support platform **2014** as noted above.

(104) As shown in the illustrative embodiment of FIGS. **16-18**, support rails **2018**, **2020** of the frame **2015** are disposed at respective right and left lateral sides **50**, **52** of patient support **2013** in spaced apart relation to each other. Each support rail **2018**, **2020** illustratively includes a torso rail **2054** and a leg rail **2056**. Each torso rail **2054** illustratively extends from the head end **30** to the mid-section **32** of the surgical support **2012**. The torso rails **2054** are each illustratively embodied as straight rails extending in parallel spaced apart relation to each other. The torso rails **2054** are illustratively connected to opposite lateral ends of beam **2022** as shown in FIG. **16**. Torso rails **2054** on each lateral side **50**, **52** illustratively connect to one leg rail **2056** on the corresponding lateral side **50**, **52** at the mid-section **32** of patient support **2013**. In the illustrative embodiment, torso rails **2054** are connected to their respective leg rails **2056** by rigid connections such that rails **2054**, **2056** do not move relative to each other.

(105) Each leg rail **2056** illustratively extends from the mid-section **32** to the foot end **34** of patient support **2013** as shown in FIGS. **17** and **18**. Each leg rail **2056** illustratively connects to one corresponding torso rail **2056** at the mid-section **32** of patient support **2013**. Each leg rail **2056** illustratively includes a first sub-rail **2058** and a second sub-rail **2062** as shown in FIGS. **17** and **18**. In the illustrative embodiment shown in FIG. **18**, first sub-rail **2058** of first rail **18** extends from mid-section **32** toward foot end **34** at angle α relative to its corresponding torso rail **2054** of the same first support rail **2018** (the position of the torso rail **2054** indicated by dotted line **35** in FIG. **18**). In the illustrative embodiment, the first sub-rail **2058** is straight and extends at angle α of about 35 degrees relative to its corresponding torso rail **2054** of first support rail **2018**.

(106) As illustratively suggested in FIGS. **17** and **18**, the angle α of each first sub-rail **2058** is downward relative to their respective torso rails **2054**, however, the indication of the relative direction downward is descriptive and is not intended to limit the orientation of the frame **2015** of the surgical support **2000**. In some embodiments, the first sub-rail **2058** of each first and second support rails **2018**, **2020** may have any angle α relative to its corresponding torso rail **2054** including but not limited to any angle within the range of about −15 to about 90 degrees, for example.

(107) As shown in FIG. **17**, support platform **2014** illustratively includes the torso platform **2036** and the leg platform **2038**. Torso platform **2036** extends from head end **30** to mid-section **32** of patient support **2013**. Leg platform **2038** extends from the mid-section **32** to a foot end **2042** near

the foot end **34** of the patient support **2013**.

(108) Leg platform **2038** is hingedly supported by frame **2015** to pivot about an axis **25** extending laterally relative to patient support **2012** such that a foot end **2042** of leg platform **2038** is lowered relative to its head end **2040** to provide leg break to an occupying patient as shown in FIG. **17**. In the illustrative embodiment shown in FIGS. **16-18**, leg platform **2038** is supported by the actuator assembly **2016** so as to be cantilevered with respect to the hinged connection to torso platform **2036**. Head end **2040** is hingedly connected to frame **2015** in some embodiments, but regardless of whether head end **2040** is hingedly connected to torso platform **2036** or frame **2015**, foot end **2042** of leg platform **2038** is a free end having no direct connection with any support structure, for example, foot end **2042** illustratively has no direct structural connection to frame **2015**, bracket **2017**, and/or tower **21**.

(109) In the illustrative embodiment shown in FIGS. **17** and **18**, the protection sheath **2070** is illustratively disposed near the foot end **34** of the surgical support **2000** to provide pinch protection during movement of the leg platform **2038**. Protection sheath **2070** is illustratively coupled to each of the second sub-rails **2062** of each leg rail **2056** of each of the first and second support rails **2018**, **2020**. In the illustrative embodiment, the protection sheath **2070** extends across the space defined between the second sub-rails **2062** of each of the first and second support rails **2018**, **2020**.

(110) In the illustrative embodiment shown in FIGS. **19** and **20**, the protection sheath **2070** is embodied as a shovel-shaped guard including a tray **2072** extending between and connecting to a pair of arms **2074**. Tray **2072** illustratively includes a front side **2079** having a guide surface **2078** disposed thereon and having a shape that corresponds closely to the shape and the travel path of the leg platform **2038** to prevent pinch points during movement of the leg platform **2038**. In the illustrative embodiment, the guide surface **2078** includes a curvature C.sub.1 along the vertical direction (in the orientation shown in FIG. **19**) corresponding closely to the travel path of the leg platform **2058** and a curvature C.sub.2 along horizontal direction (in the orientation shown in FIG. **19**) corresponding closely to the shape of the foot end **2042** of the leg platform **2038**. By reducing spacing between the frame **2015** and the leg platform **2038** using the protection sheath **2070**, the potential for a portion of a patient's, surgeon's or other person's body to be pinched between parts of the surgical support **2000** is reduced.

(111) As best shown in FIG. **19**, each of the arms **2074** defines an opening **2076** and a cavity **2077** extending from the opening **2076** for receiving one of the second sub-rails **2062** for connection between the protection sheath **2070** and the frame **2015**. Arms **2074** each have a tapered width extending between a thicker width proximate to a top edge **2082** and a thinner width proximate to the opening **2076**. Each arm **2074** illustratively includes a rounded front edge **2086** for comfortable contact with a patient supported by the surgical support **2000**. In the illustrative embodiment, the arms **2074** are arranged in spaced apart relation to each other to define a gap **2088** therebetween for receiving passage of foot end **2042** of the leg platform **2038** in close proximity to the arms **2074** and the tray **2072** during movement of the leg platform **2038** to reduce pinch points.

(112) As shown in FIG. **20**, the protection sheath **2070** illustratively includes an opening **2080** formed on a rear side **2081** thereof near a top edge **2082** of the sheath **2070** and a cavity **2084** extending from the opening **2080** into the sheath **2070** for receiving the beam **2024**. The opening **2080** extends between each of the arms **2074** along the top edge **2082** and the cavity **2084** is configured to receive the beam **2024** arranged proximate to the foot end **34** of the frame **2015** for connection with the support bracket **2017** through the opening **2080** via a floating arm **44** as shown in FIG. **16**. In the illustrative embodiment, each of the cavities **2077** of the arms **2074** communicate with the cavity **2084** of the rear side **2081** of the protection sheath **2070** to from a continuous pathway such that the frame **2015** near the foot end **34**, including the second sub-rails **2062** while connected to the beam **2024**, is received within the sheath **2070** to reduce pinch points during movement of the leg platform **2038**.

(113) In the illustrative embodiment, the protection sheath **2070** is embodied as a hollow shell

formed of plastic. In some embodiments, the protection sheath **2070** may be formed with any suitable interior structure and/or with any suitable materials. In the illustrative embodiment, divots or depressions **2089** are formed in rear side **2081** of sheath **2070** and extend toward tray **2072** so as to help rigidify tray **2072**. That is, if tray **2072** flexes or attempts to flex toward rear side **2081**, contact with depressions **2089** limits the amount of flexion that can occur.
(114) Although certain illustrative embodiments have been described in detail above, variations and modifications exist within the scope and spirit of this disclosure as described and as defined in the following claims.

Claims

1. A surgical patient support system, comprising: a base frame including a head elevator tower and a foot elevator tower each having a support bracket connected thereto for movement of the support brackets between higher and lower positions as the head and foot elevator towers are raised and lowered, respectively, a support frame including first and second rails extending parallel to each other from a head end to a foot end, a head-cross beam and a foot-cross beam extending between the first and second rails at the head end and foot end respectively, and connection arms including a head-connection arm engaged with the head-cross beam and coupled with the support bracket of the head tower and a leg-connection arm engaged with the leg-cross beam and coupled with the support bracket of the leg tower, a support platform coupled to the support frame and including a torso section and a leg section, and an actuator coupled to the support frame and configured to support the leg section, wherein the actuator is operated electrically to move the leg section relative to the torso section between a raised position and a lowered position to create a leg break of a surgical patient in a lateral position, wherein the support frame comprises a first support frame and further comprising a second support frame coupled to the support brackets and arranged perpendicular to the support platform when the leg section is in the raised position.
2. The surgical patient support of claim 1, wherein the leg section of the support platform is hingedly attached to the support frame to move between the raised position and the lowered position and the actuator is pivotably connected to the leg section of the platform on a bottom side thereof.
3. The surgical patient support of claim 1, wherein the actuator is configured for operation between an extended position and a retracted position, the extended position of the actuator corresponds to the raised position of the leg section, and the retracted position of the actuator corresponds to the lowered position of the leg section.
4. The surgical patient support of claim 3, wherein the lowered position is arranged to contribute about 25° of leg break to the surgical patient in the lateral position.
5. The surgical patient support of claim 3, wherein the raised position is arranged to contribute about 0° of leg break to the surgical patient in the lateral position.
6. The surgical patient support of claim 3, wherein the actuator includes a linear actuator configured to rotate an axle.
7. The surgical patient support of claim 1, wherein the first and second rails each include a torso rail which extends from the head end towards the foot end and the first and second rails define a constant width between the torso rails along the extension direction.
8. The surgical patient support of claim 1, wherein the first and second rails each include a torso rail and a leg rail, the torso rails each extending from the head-cross beam towards the foot end to connect with the leg rail of the respective support rail, and each leg rail extends from connection with the torso rail of the respective support rail towards the foot end.
9. The surgical patient support of claim 1, wherein the torso section comprises a first flat panel and the leg section comprises a second flat panel.
10. A surgical patient support system, comprising: a base frame including a head elevator tower and

a foot elevator tower each having a support bracket connected thereto for movement of the support brackets between higher and lower positions as the head and foot elevator towers are raised and lowered, respectively, a support frame including first and second rails extending parallel to each other from a head end to a foot end, a head-cross beam and a foot-cross beam extending between the first and second rails at the head end and foot end respectively, and connection arms including a head-connection arm engaged with the head-cross beam and coupled with the support bracket of the head tower and a leg-connection arm engaged with the leg-cross beam and coupled with the support bracket of the leg tower, a support platform coupled to the support frame and including a torso section and a leg section, and an actuator coupled to the support frame and configured to support the leg section, wherein the actuator is operated electrically to move the leg section relative to the torso section between a raised position and a lowered position to create a leg break of a surgical patient in a lateral position, wherein the first and second rails each include a torso rail and a leg rail, the torso rails each extending from the head-cross beam towards the foot end to connect with the leg rail of the respective support rail, and each leg rail extends from connection with the torso rail of the respective support rail towards the foot end, wherein each leg rail includes a first sub-rail and a second sub-rail, and each first sub-rail extends from connection with the respective torso rail towards the foot end at an angle relative to the respective torso rail.
11. The surgical patient support of claim 10, wherein each first sub-rail extends from connection with the respective torso rail towards the foot end at an angle of about 15 to about 35 degrees relative to the respective torso rail.
12. The surgical patient support of claim 10, wherein each second sub-rail extends from connection with the foot-cross beam for connection with the respective first sub-rail.
13. The surgical patient support of claim 10, wherein in the lowered position, the leg section of the platform is parallel to each first sub-rail.
14. The surgical patient support of claim 10, further comprising a protection sheath coupled the second sub-rails and configured to block against pinch point formation during movement of the leg section.
15. The surgical patient support of claim 8, wherein the support frame includes a cross link that extends between the leg rails and a cross arm extending from the cross link to support the actuator.
16. The surgical patient support of claim 15, wherein the actuator is pivotably connected to the cross arm.
17. A surgical patient support system, comprising: a base frame including a head elevator tower and a foot elevator tower each having a support bracket connected thereto for movement of the support brackets between higher and lower positions as the head and foot elevator towers are raised and lowered, respectively, a support frame including first and second rails extending parallel to each other from a head end to a foot end, a head-cross beam and a foot-cross beam extending between the first and second rails at the head end and foot end respectively, and connection arms including a head-connection arm engaged with the head-cross beam and coupled with the support bracket of the head tower and a leg-connection arm engaged with the leg-cross beam and coupled with the support bracket of the leg tower, a support platform coupled to the support frame and including a torso section and a leg section, and an actuator coupled to the support frame and configured to support the leg section, wherein the actuator is operated electrically to move the leg section relative to the torso section between a raised position and a lowered position to create a leg break of a surgical patient in a lateral position, wherein the torso section comprises a first flat panel and the leg section comprises a second flat panel, wherein the first and second rails each include a torso rail and a leg rail and wherein a bottom surface of the second flat panel is spaced apart from the leg rails when the leg section is in the raised position.
18. The surgical patient support of claim 17, wherein the bottom surface of the second flat panel rests atop the leg rails when the leg section is in the lowered position.

19. The surgical patient support of claim 17, wherein the first flat panel rests atop the torso rails during movement of the leg section between the raised and lowered positions.